

The Competing System For Two Species



Jinming Li

China

June 20, 2017

Summary

In Northern China, herdsmen in the Inner Mongolia prairie have been raising cattle and horses for thousands of years. However, due to the modernization, grasslands are becoming more and more limited. Therefore, a competing relationship between cattle and horse arises. In this scientific report, a system of differential equations is introduced to simulate the population of cattle and horse when they are put together. The long term behavior of the differential equations system would be discussed. Finally, based on the discussion, conclusions and policy recommendations are drawn. The purpose of this scientific report is to provide useful information and recommendations to herdsmen in the Inner Mongolia prairie in order to better manage the livestock farming.

Contents

Summary.....	2
List of Variables.....	4
Chapter 1.....	5
Introduction.....	5
Chapter 2.....	7
Simplified Population Model.....	7
Chapter 3.....	9
Improving Simplified Population Model.....	9
Chapter 4.....	11
An Interacting System of Population Model.....	11
Chapter 5.....	13
An General view of Population Model System.....	13

List of Variables

c : cattle in the unit of head

h : horse in the unit of head

t : time in the unit of day

α : the birth rate of cattle

β : the birth rate of horse

γ : the competition of cattle and horse

K_c : the carrying capacity of cattle in an area of 1 kilometer

K_h : the carrying capacity of horse in an area of 1 kilometer

Chapter 1

Introduction

In Northern China, the Inner Mongolia prairie covers an area of thousands of kilometers. Over thousands of years, herdsmen have been raising cattle and horses in the Inner Mongolia prairie which is an ideal place for raising livestock as for countless grass can afford cattle and horses to grow up without competing with each other. However, due to the modernization, cities have become larger and larger and hence more and more lands are transformed from grassland into urban land. This gives rise to a serious problem.

With more and more grasslands are being used to build houses and factories for the city development, herdsmen do not have countless grassland anymore. They only have limited grassland for raising cattle and horses, and because of that the competing relationship between cattle and horses rises.

In the following chapters, we want to discuss how the competing relationship between cattle and horses behaves in the long run and based on mathematical models which we set up to simulate the situation, we would be able to discuss the long term behavior of this competing relationship. The purpose of this report is that according to the models, we might provide the herdsmen with useful information for their decision making.

This scientific report is not without its limitations. All the mathematical models and conclusions should be based on the following assumptions: (1) the birth rate and death rate of cattle and horses are constant and will not change over time. (2) cattle and horses only have one food source which is grass, they do not have other food source. (3) cattle and horses do not hurt each other, that is to say they only compete for grass but not hurting others.

(4) cattle and horses are the only two species exist in the Inner Mongolia prairie, other species do not exist or have no effect on these two species.

In this scientific report, a system of differential equations simulating the cattle and horses populations would be proposed and discussion for long term behavior would base on the differential equations.

In Chapter 2, a simplified population model would be built which we do not let these two species interact while at the same time, no limited place for growing. By doing this, readers can have a close look at how cattle or horse population behaves when two species are separated under free condition.

In Chapter 3. an improvement would be proposed in which a carrying capacity is added. By doing this, we simulate the cattle and horse population separately but in limited place as this might be in accord with the reality.

In Chapter 4, an interaction would be put into effect. By doing this, the readers can have a better understanding of what is going to happen when cattle and horses are put together in the long run.

In Chapter 5, a general perspective would be given to discuss how initial condition might have a different results to the system of differential equations. By doing this, readers might be able to look into the equilibrium points, the vector field and the phase plane to discuss the initial condition effects.

Finally, any recommendation for better management would be proposed based on the discussions we have in the previous chapters.

Chapter 2

Simplified Population Model

In this chapter, a simplified population model is introduced in which cattle and horses are separate as well as countless grass would be provided. By doing this, readers might have a clear view of how cattle and horses populations behave when separated under free condition.

According to “Malthusian Theory of Population Growth”, the population growth rate is proportional to the population. Here, we use “c” to refer to cattle, “h” to refer to horses. The unit of cattle and horses are head, the unit for the time t is day. Therefore, we want to simulate the growth rate of cattle and horses in days.

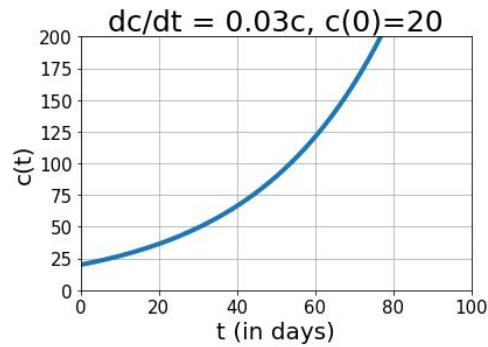
The differential equation to simulate the Malthusian population growth model: $\frac{dc}{dt} = \alpha c$ with α, β are constants, and the initial value $c_0=20, h_0=10$.

$$\frac{dh}{dt} = \beta h$$

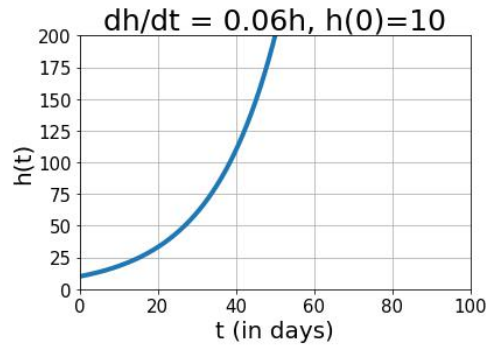
Here, we make a reasonable guess, the birth rate of cattle and horses are each 3% and 6%, the death rate are each 0.02% and 0.04% which are small, so here the death rates are ignored . Therefore, the differential equation can be written as: $\frac{dc}{dt} = 0.03 c$ with the initial value $c_0=20, h_0=10$.

$$\frac{dh}{dt} = 0.06 h$$

Choosing the time step as 0.1 for the estimate error is acceptable (roughly 0.5 compared to the time step as 0.005 at 80 days), a solution graph is proposed. With the initial value of cattle at 20, the graph of the solution to the differential equation is given below. After 80 days, the population of cattle reaches 220.



Similarly, choosing the time step as 0.1 for the horses population (estimate error is roughly 8.6 compared to the time step as 0.005 at 80 days), with the initial value of horses at 10, the graph of the solution to the differential equation is given below. After 80 days, the population of cattle reaches 1198.



Above all, a rough graph of the Malthusian population grow model states that under free condition, a specie might grow exponentially. We can solve these two differential equations quantitatively and we get the solution:
 $c(t) = e^{0.03t}, h(t) = e^{0.06t}$.

Chapter 3

Improving Simplified Population Model

In this chapter, an improvement to the simplified population model is made. With more and more grasslands are transformed urban lands, the food resources for cattle and horses are not longer unlimited. To simplify the situation, we suppose the living environment for cattle and horses is restricted to 1 kilometer.

According to the “Logistic Growth Model”, a carrying capacity “K” is introduced which refers to the biggest number of population an environment can hold. Based on the assumptions and Logistic growth model we have, the differential equations of cattle and horses under limited food resources can be written as:

$$\begin{aligned}\frac{dc}{dt} &= \alpha c \left(1 - \frac{c}{K_c}\right) \\ \frac{dh}{dt} &= \beta c \left(1 - \frac{h}{K_h}\right)\end{aligned}$$

with c, h stand for the population

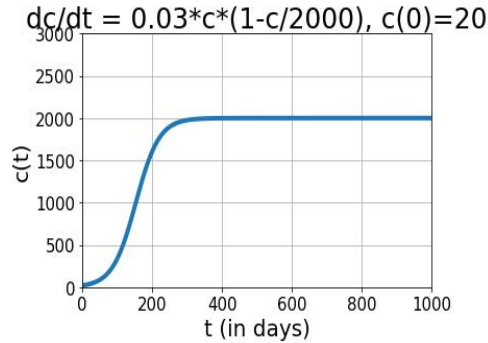
of cattle and horse, α and β stand for the birth rate of cattle and horse, K_c and K_h stand for the carrying capacity of cattle and horse. We suppose the carrying capacity of cattle in a area of 1 kilometer is 2000, and that of horse is 1500. Therefore, with the information, the differential equations can be written as:

$$\begin{aligned}\frac{dc}{dt} &= 0.03 c \left(1 - \frac{c}{2000}\right) \\ \frac{dh}{dt} &= 0.06 c \left(1 - \frac{h}{1500}\right)\end{aligned}$$

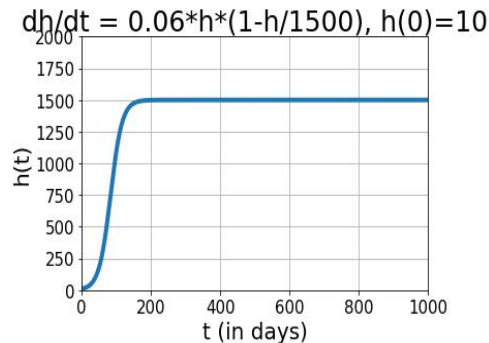
value $c_0=20$, $h_0=10$.

Choosing the time step as 0.01 for the estimate error is acceptable (roughly 0.003 compared to the time step as 0.005 at 300 days), a solution graph is

proposed. With the initial value of cattle at 20, the graph of the solution to the differential equation is given below. After 300 days, the population of cattle reaches 1975.



Choosing the time step as 0.01 for the estimate error is acceptable (roughly 0.004 compared to the time step as 0.005 at 200 days), a solution graph is proposed. With the initial value of horses at 10, the graph of the solution to the differential equation is given below. After 200 days, the population of cattle reaches 1498.



Above all, a logistic growth model is introduced in which the limited resource results in the carrying capacity for an environment. This is more realistic to the situation in Inner Mongolia prairie.

From the solution graph of the two differential equations, for cattle, the birth rate is lower than horses which results in a flatter curve, and finally gets asymptotic to 2000. As for horses, the sheerer curve shows the higher birth rate and finally gets asymptotic to 1500.

Chapter 4

An Interacting System of Population Model

In this chapter, a more realistic improvement is made to adapt the population model. With grasslands are limited and the carrying capacity is introduced, an interacting system of population model would come into use when cattle and horses are put together. Therefore, a competing relationship arises. Based on the assumptions we have in the previous chapters, cattle and horses only compete for grass. By putting cattle and horses into a system, an interaction might be seen, hence we can make a discussion of the long term behavior of this competing system.

The population growth rate of each group is determined not only by the birth rate but also by the competitiveness. Here, we define the competition as proportional to the population of each group. Because of the competition, each specie might not get enough food they need and hence part of the specie dies. Therefore, the effect of competition is negative. The system of differential equations can be written as:

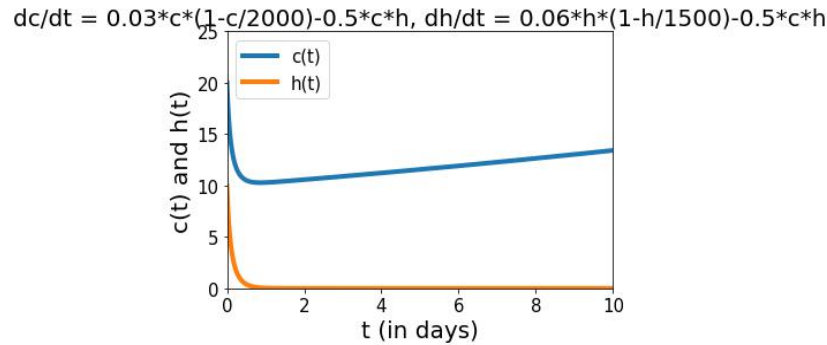
$$\begin{cases} \frac{dc}{dt} = \alpha c \left(1 - \frac{c}{K_c}\right) - \gamma ch \\ \frac{dh}{dt} = \beta c \left(1 - \frac{h}{K_h}\right) - \gamma ch \end{cases}, (c_0, h_0) = (20, 10)$$

The competitiveness is difficult to measure, so we make a guess and assume $\gamma=0.5$. With the information given in

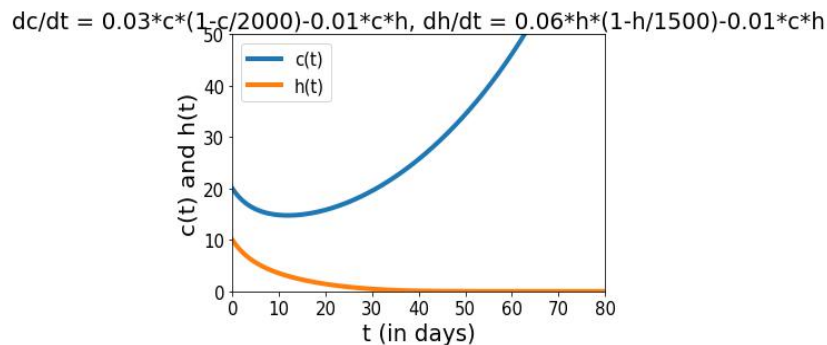
the previous chapters, the system of differential equations can be written as:

$$\begin{cases} \frac{dc}{dt} = 0.03 c \left(1 - \frac{c}{2000}\right) - 0.5 ch \\ \frac{dh}{dt} = 0.06 c \left(1 - \frac{h}{1500}\right) - 0.5 ch \end{cases}, (c_0, h_0) = (20, 10)$$

Choosing the time step as 0.01 for the estimate error is acceptable (roughly 0.3 for cattle and 0.003 for horses compared to the time step as 0.005 at 2 days), a solution graph is proposed. With the initial value of cattle at 20, horses at 10, the graph of the solution to the differential equation is given below.



However, the result is not realistic. For horse population, it dies out with 1 day which is impossible. The measurement of competitiveness γ is not too high. Therefore, we change $\gamma=0.01$. The solution graph of the differential equations system is shown below:



The result is more realistic. With the initial value $c_0=20$, $h_0=10$, the population of horse dies out in 40 days. The population of cattle goes down at the first 10 days, and then starts to increase until it gets asymptotic to the carrying capacity 2000

Chapter 5

An General view of Population Model System

To Be Continued...